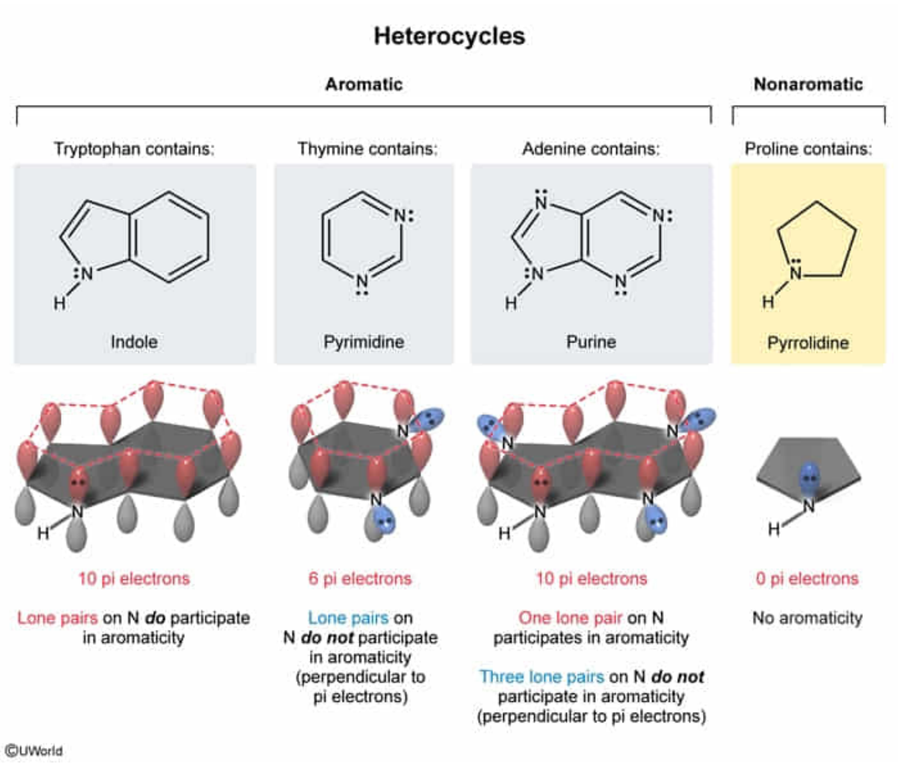
All of the following can be described as an aromatic heterocycle EXCEPT:

- A. tryptophan. [12%]
- B. proline. [69%]
- C. thymine. [9%]
- D. adenine. [9%]

Correct
68% answered correctly

Explanation:



Aromatic heterocycles are commonly found in biological systems. A heterocycle is any cyclic compound containing at least one heteroatom (ie, an atom other than carbon or hydrogen). To be classified as **aromatic**, a compound must have:

1. **Conjugated pi bonds** in a **cyclic** structure

2. **Unhybridized *p* orbitals** present in each atom in the ring

3. Planar geometry, forming a continuous ring of parallel, overlapping unhybridized *p* orbitals

4. **4*n* + 2 pi electrons** (Hückel rule), where n is a non-negative integer

Tryptophan, thymine, and adenine are all considered aromatic heterocycles because they have nitrogen-containing rings (indole, pyrimidine, and purine, respectively) with conjugated pi bonds and overlapping

unhybridized *p* orbitals **(Choices A, C, and D)**. Although thymine may not appear to have conjugated pi bonds, a resonance structure can be drawn to show a conjugated system similar to pyrimidine. Basic nitrogen atoms, such as in pyrimidine and purine, have a double bond placing the lone pair electrons in an sp^2 orbital perpendicular to the conjugated *p* orbitals containing the pi electrons. Electrons in orbitals perpendicular to the conjugated plane cannot undergo side-to-side overlap with the *p* orbitals and therefore *do not* participate in aromaticity. Nonbasic nitrogen atoms, such as in purine and indole, have three bonds, putting the lone pair of electrons into a *p* orbital parallel to other pi electron-containing *p* orbitals.

Parallel orbitals permit side-to-side overlap; therefore, these electrons *do* participate in aromaticity. As a result, thymine has 6 pi electrons, and tryptophan and adenine each have 10 pi electrons; all satisfy Hückel rule.

(Choice B) Proline can be classified as a heterocycle, but it is not considered aromatic because it is not conjugated and does not contain any pi electrons.

Educational objective:
To be considered aromatic, a compound must be planar and cyclic with conjugated pi bonds. Each atom must have parallel, overlapping unhybridized *p* orbitals. Finally, an aromatic compound will satisfy Hückel rule and have $4n + 2$ pi electrons.

Time spent:0 sec

Subject: Organic Chemistry

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System: Functional Groups and Their Reactions